SET OPERATOINS

In [1]:
```
A={1,2,3,4,5}
B={4,5,6,7,8}
C={8,9,10}
```

In [2]:
```
A|B|C
```

Out[2]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

In [3]:
```
A.union(B)
```

Out[3]: {1, 2, 3, 4, 5, 6, 7, 8}

In [4]:
```
B.union(C)
```

Out[4]: {4, 5, 6, 7, 8, 9, 10}

In [5]:
```
print(A)
print(B)
print(C)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

In [6]:
```
A.intersection(B)
```

Out[6]: {4, 5}

In [7]:
```
B.intersection(C)
```

Out[7]: {8}

In [9]:
```
A.intersection(C)
```

Out[9]: set()

In [10]:
```
A&B
```

Out[10]: {4, 5}

In [11]:
```
B&C
```

Out[11]: {8}

In [12]:
```
A.difference(B)
```

Out[12]: {1, 2, 3}

In [13]:
```
A-B
```

Out[13]: {1, 2, 3}

In [14]:
```
A-C
```
Out[14]: {1, 2, 3, 4, 5}

In [15]:
```
C-A
```
Out[15]: {8, 9, 10}

In [16]:
```
A.symmetric_difference(B)
```
Out[16]: {1, 2, 3, 6, 7, 8}

In [17]:
```
A^B
```
Out[17]: {1, 2, 3, 6, 7, 8}

In [18]:
```
A.symmetric_difference_update(B)
```
In [19]:
```
A
```
Out[19]: {1, 2, 3, 6, 7, 8}

superset, subset and disjoint

In [1]:
```
A = {1,2,3,4,5,6,7,8,9}
B = {3,4,5,6,7,8}
C = {10,20,30,40}
```
In [2]:
```
B.issubset(A)
```
Out[2]: True

In [3]:
```
A.issuperset(B)
```
Out[3]: True

In [4]:
```
C.isdisjoint(B)
```
Out[4]: True

In [5]:
```
C.isdisjoint(A)
```
Out[5]: True

In [ ]: